

Market Research & Research Synthesis

NZ GORHAM

DESI 2701

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THANK YOU

Problem/Opportunity statements

1. College students need a way to communicate with TAs and Professors because students need help with HW and sometimes aren't able to go to in-person or ZOOM OHs. This can also reduce the stress of students not getting the help they need
2. Professors/TAs need to have empathy because it builds strong connections with students and hopefully learn how to improve the class
3. College students can benefit from having an AI teaching website doesn't get out the answers to HW problems because it can still help students learn on the materials in the class (reduce stress), but cheating is not the easy way to through college
4. Classes need to be fair and considerate because if it doesn't take the chance to learn or make it fair to all the students, then it will stress out the students
5. Professors can benefit from making classes more engaging because students can participate and can fun out the class and out have stress for the class
6. Disabled students can benefit from having DRC accommodations because it helped them to succeed in college and not be stressed out of assignment and exam time (they can get extended time)
7. Disabled students need medication for their disabilities because it can help them to focus and maintain a good healthy body
8. Students need a better way to limit distractions because the distractions can cause stress on them and not get work done
9. Professors/TAs can benefit from not being harsh graders because when students try their best on assignments and exams and Professors/TAs give them a 0 on the problem, that will cause stress on the students. It's like if the Professors/TAs are being Tiger Mothers
10. Students need to attend classes because being behind on class is not fun and can lead to stress of catching up
11. Professors need to post their lectures notes on Canvas because students and TAs can look at them when they aren't at class and use them as reference

FINAL PICK (EDIT)

College students need a stress free way to communicate with TAs and Professors because not getting help is stressful, but often talking to them is scary.

WHY I CHOSE THIS

Sharing my statements with my partner, she liked my first idea a lot and suggested that I combine 1 with 2 and 3 because all of them are related and she is dealing with a professor that isn't nice. However, looking at this more, I decided to make one more edit to this so it can work precisely well for what I want to solve (thanks to Beth and Nefertari).

AEIOU context analysis (Activities)

ACTIVITIES

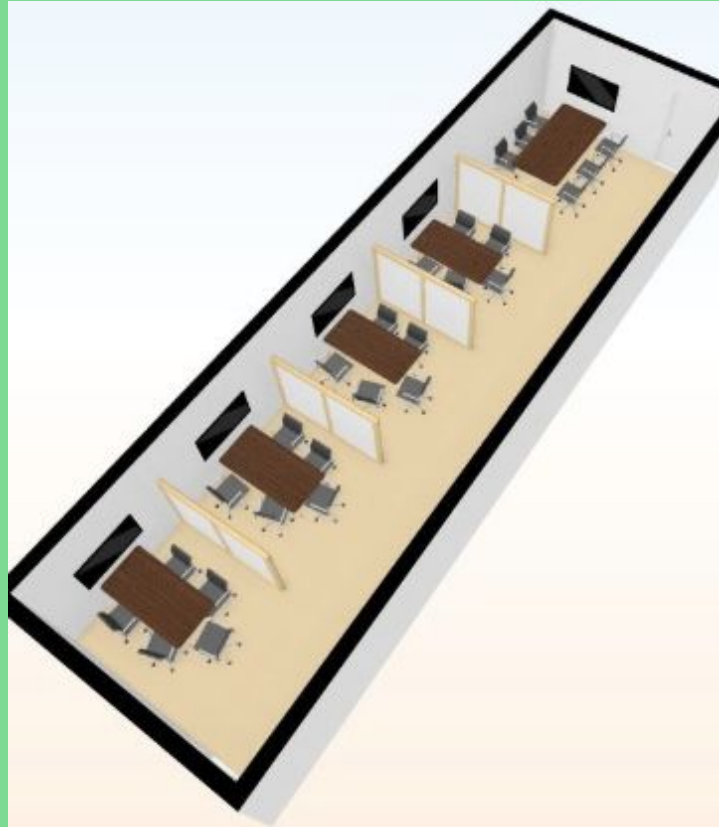
- Students studying
- Students doing homework
- Students socializing
- Office Hours being hosted by TAs (and Professors)
- Students working projects/labs
- Watching something on the TVs
- Students doing research



AEIOU context analysis (Environment(s))

ENVIRONMENT(S)

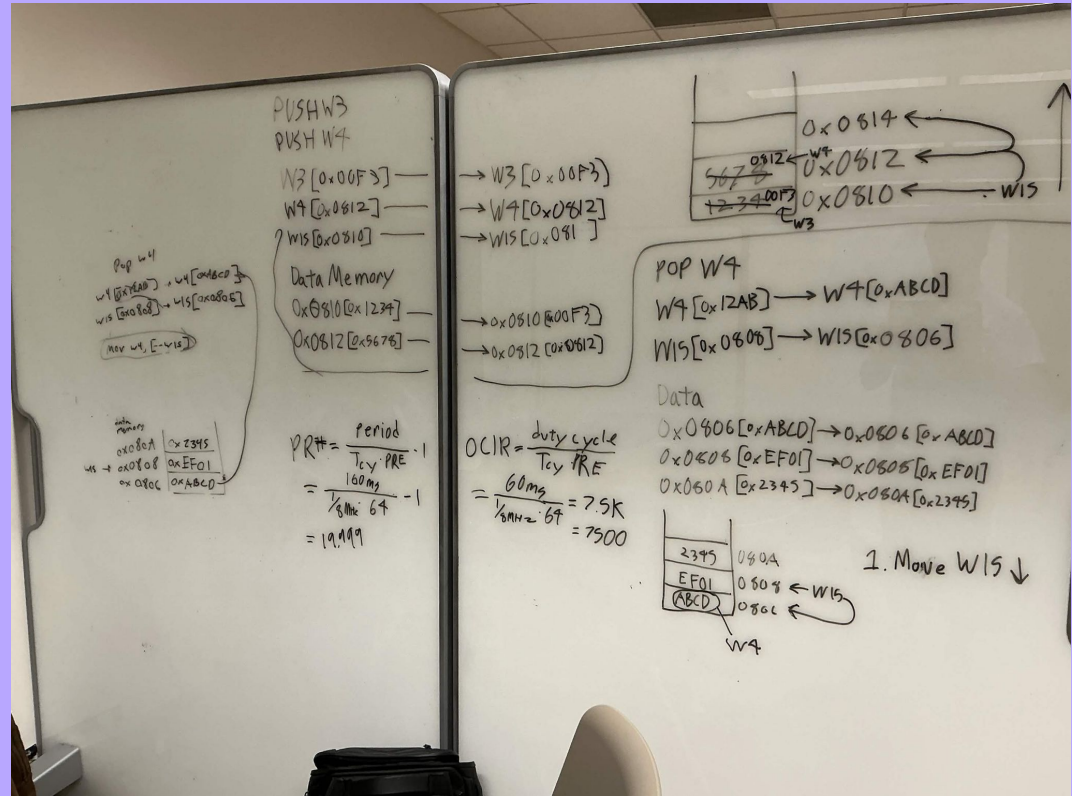
- Classroom
- Socialization
- Learning
- Quiet
- Pods
- Loud
- Crowded
- Learning Center (A room to learn)



AEIOU context analysis (Interactions)

INTERACTIONS

- Students asking questions to the TA
- Student and TA going over problems step by step or understanding why they [the student] got it wrong
- Students talking to other students
- Student learning about the content of the course
- Small Groups
- Big Groups
- TA conversations



AEIOU context analysis (Objects)

OBJECTS

- Whiteboards
- Chairs
- Tables
- Markers
- Erasers
- TV
- Outlets
- Flyers
- Physical Textbooks
- Book Shelf
- Kiosk (Students can make reservations to do a study group, work on a project, or anything else)



AEIOU context analysis (Users)

USERS

- ECE Students
- Non ECE Students
- Undergraduate Teaching Assistants
- Graduate Teaching Assistants
- Professors
- Members of Club
- Executive Members of Club
- ECE Staff
- Researchers



Market research table (1-5)

PICTURE					
NAME	ZOOM OFFICE HOURS	DISCORD	GROUP ME	PIAZZA	AI
DESCRIPTION KEY FEATURES	An online call where people can host online social events, private online meetings, individual calls, having office hours so students can get the help they need even if they can't make it to in-person Office Hours.	An app/website where people can communicate and socialize. This one is very popular.	An app/website where people can communicate and socialize. (From how I used it, it feels more towards college dormrooms and athletics).	An app/website where people can communicate and socialize. (From how I used it, it feels more towards college classes only).	An app/website where people can use ChatGPT, Deekseek, other sites to get help understanding topics, finding solutions, or solving any problem you have.
PROS	• Time saver • Easier access for help and to speak with a TA or Professor	• Students communicating with all students, TAs, and the professor at once. • Saves the students time from going to OHs (in-person)	• Students communicating with all students, TAs, and the professor at once. • Saves the students time from going to OHs (in-person)	• Students communicating with all students, TAs, and the professor at once. • Saves the students time from going to OHs (in-person)	• Easy access • Time saver • Give students the solution • Help them with homework problems (give them hints)
CONS	• No [students] show up • Zoom glitches and crashes	• Students spamming • Saying/displaying inappropriate messages/videos • Cyberbully anyone	• Students spamming • Saying/displaying inappropriate messages/videos • Cyberbully anyone	• Students spamming • Saying/displaying inappropriate messages/videos • Cyberbully anyone	• Professors and departments not accepting it • Unethical way for students to learn and develop

Market research table (6-10)

PICTURE					
NAME	IN PERSON OFFICE HOURS	STUDY GROUP	1:1 OFFICE HOURS	EMAIL	CANVAS DISCUSSIONS
DESCRIPTION KEY FEATURES	Students meeting with the TA or Professor in-person in a learning center room or office room to get help with homework, exams, or anything else.	A group of students doing homework or studying for exams. This can include the TA as well as long as it's in a learning room, lab room, or classroom.	Private lesson: Student getting time with TA and professor.	An app/website where people can communicate. Students can email TAs or professors to ask for help or to schedule private office hours with them.	Discussion boards can be made in Canvas so students can ask questions on how to solve problems (Calc 1 & 2) or get feedback on assignments (DESI 2701). Everyone can reply.
PROS	- Students having better interactions with the professor or TA - Ask questions without waiting for a long time	- Working with people that you are happy that you are happy with - Time saver (as in like if one person got the answer, then everyone can get the answer correct) - Ethical way for students to build their teamwork skills	- Student can learn easier - Build a strong connection between student and TA/Professor (as long as it goes well)	Students not needing to worry about what others would say or think behind their back	- Students/TAs/Professors giving good feedback - Students getting a ton of feedback
CONS	- No [students] show up, make the learning center room crowded - Students worrying about what others would say or think behind their back	- Too many distractions - Professors not approving it - Sometimes an unethical way for students to learn and develop (are some of the members really learning or just wanting answers and skip the learning process)	- Arguments - Students worrying about what the professor or TA would reply	- No reply or long delay with replying - Students worrying about what the professor or TA would reply	- No reply or long delay with replying - Students/TAs/Professors giving poor feedback

2x2 & Reflection



Summary:

After creating this graph, I notice that the majority of my 10 things are online. While that isn't a bad thing, some students (like me) prefer in-person.

New opportunity/Solution:

As a TA, one new opportunity I would consider is having more in-person office hours and allow study groups. This can help build trust and connections between students and TAs/professors, students can get their questions answered more efficiently, and students can work with TAs they can connect with well.

2x2 & Reflection



Summary:

After creating this graph, I will admit, this is based on my experience (there is bias :(...)). I will say this though, I think that we all do get annoyed we ask questions and we get no response.

New opportunity/Solution:

As a student, I would like to solve the issue on students not getting responses or being responses super late. I would create an extension for gmail so when TAs get an email from a student, they can use color coordination to communicate if the TA won't be able to respond right away. Another one (while no fully necessary) is to create an AI website so students don't fully have to rely on it as their TA or professor. If Professor approves, then fine. At the time, AI should be used fairly. Plus, it takes away an opportunity for students to build connections with TAs and professors

4 people for brainstorming

MEETING:
Zoom on 3/27 @ 3pm



PERSON 1:

Name: Maryam Jama

School: University of Minnesota

Grade: Freshman

Major: User Experience Design

Relation to me: Student in this class

Relation to this topic: Maryam wants to get help for her design class (DESI 1311) but her professor is harsh



PERSON 2:

Name: Dublin Flynn

School: University of Minnesota

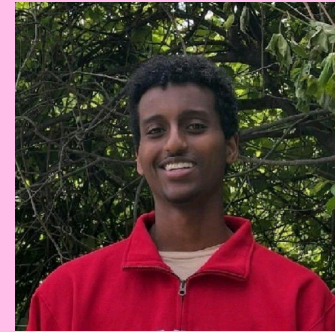
Grade: Junior

Major: Computer Engineering

Relation to me:

Roommate/Friend

Relation to this topic: Dublin wanted his 2015 professor to not say negative comments so he can't feel down on himself



PERSON 3:

Name: Ezana Medlin

School: University of Wisconsin

Grade: Junior

Major: Business

Relation to me: Friend

Relation to this topic: While Ezana didn't have 100% terrible TAs or Professors, he would like for both to be patient, set ground rules early and firm, and offer flexibility



Thank you!